

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026
Biology — Class IX | Paper: SSC-I
Syllabus: Chapter 7 — Metabolism

Total Marks: 65

Time: 2h 30m

Version: 1

General Instructions:

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The sum of all chemical reactions in a living cell is called:	☐ Catabolism	☐ Anabolism	☐ Metabolism	☐ Enzymology	
2	Enzymes are biologically active globular proteins produced by:	☐ Dead cells	☐ Living cells	☐ Inorganic ions	☐ Ribosomes only	
3	The part of an enzyme where substrate binds and reaction takes place is the:	☐ Allosteric site	☐ Prosthetic group	☐ Active site	☐ Coenzyme	
4	At very high temperature, an enzyme loses function because its structure is:	☐ Activated	☐ Denatured	☐ Phosphorylated	☐ Synthesised	
5	In competitive inhibition, the inhibitor blocks enzyme activity by binding to the:	☐ Allosteric site	☐ Substrate	☐ Active site	☐ Cofactor	
6	ATP is called the “energy currency” of the cell because it:	☐ Is in the nucleus	☐ Stores and releases energy	☐ Builds proteins	☐ Carries oxygen	
7	ATP molecule consists of adenine, ribose sugar and:	☐ Two phosphate groups	☐ Three phosphate groups	☐ One phosphate group	☐ Four phosphate groups	
8	During photosynthesis, oxygen is released as a by-product through:	☐ Calvin cycle	☐ Photolysis of water	☐ Glycolysis	☐ Krebs cycle	
9	Light-independent reactions of photosynthesis (Calvin cycle) occur in the:	☐ Thylakoid membrane	☐ Cytoplasm	☐ Stroma of chloroplast	☐ Mitochondrial matrix	
10	Glycolysis produces 2 pyruvic acid from glucose with a net gain of:	☐ 36 ATP	☐ 4 ATP	☐ 2 ATP	☐ 38 ATP	
11	Anaerobic respiration in yeast (alcoholic fermentation) produces ethyl alcohol and:	☐ Water	☐ Lactic acid	☐ Carbon dioxide	☐ Oxygen	
12	The final electron acceptor in the electron transport chain (ETC) is:	☐ NADH	☐ Carbon dioxide	☐ Oxygen	☐ Water	

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Define metabolism. Differentiate between catabolism and anabolism with one example each.	3	OR	Why are enzymes called biological catalysts? Give two characteristics that support this.	3
2(ii)	Name and describe three types of cofactors with one example each.	3	OR	Explain the term “enzymes are specific.” Give one example to support your answer.	3
2(iii)	Describe the mechanism of enzyme action with reference to ES complex and EP complex.	3	OR	What is energy of activation? How do enzymes lower activation energy?	3
2(iv)	How does temperature affect enzyme activity? What happens at very high and very low temperatures?	3	OR	How does pH affect enzyme activity? Give optimum pH values for pepsin and trypsin.	3
2(v)	Distinguish between competitive inhibition and non-competitive inhibition of enzymes.	3	OR	What is the allosteric site? How does a non-competitive inhibitor damage the active site?	3
2(vi)	Describe the structure of an ATP molecule. Name its three components.	3	OR	Explain the ATP-ADP cycle. How is energy stored and released in this cycle?	3
2(vii)	Write the overall equation for photosynthesis. Identify the raw materials and products.	3	OR	Name three limiting factors of photosynthesis and briefly explain how each affects the rate.	3
2(viii)	What are light-dependent reactions of photosynthesis? Where do they occur and what are the products?	3	OR	Describe photolysis of water. What is its role in the light-dependent reactions?	3
2(ix)	Briefly describe the four stages of aerobic respiration in the correct order.	3	OR	Write the equation for aerobic respiration and state how many ATP molecules are produced from one glucose molecule.	3
2(x)	What is glycolysis? Where does it occur and what are its products?	3	OR	What is the role of the Krebs cycle in aerobic respiration? Name the products formed.	3
2(xi)	Compare alcoholic fermentation and lactic acid fermentation. Give one biological example of each.	3	OR	Give four reasons why anaerobic respiration is important for living organisms.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Define enzyme. Describe any five characteristics of enzymes with examples where appropriate.	5	OR	Explain how three environmental factors — temperature, pH, and substrate concentration — affect enzyme activity. Include denaturation and saturation.	5
3(ii)	Describe the mechanism of enzyme action. Explain how an enzyme lowers activation energy and distinguish the lock-and-key concept from the induced-fit model.	5	OR	Describe competitive and non-competitive inhibition in detail. What is the role of the allosteric site? Give examples of inhibitors from daily life.	5
3(iii)	Explain the role of ATP as the energy currency of the cell. Describe the structure of ATP, the ATP-ADP cycle, and where ATP is used in the body.	5	OR	Describe the process of photosynthesis. Write the overall equation. Explain the light-dependent reactions and the Calvin cycle (light-independent reactions) in sequence.	5
3(iv)	Compare aerobic and anaerobic respiration. Describe glycolysis as the common first stage and explain what happens to pyruvic acid in each pathway. Include relevant equations.	5	OR	Describe the mechanism of aerobic respiration in four stages: glycolysis, formation of Acetyl CoA, Krebs cycle, and electron transport chain. State the ATP yield at each stage.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 7 — Metabolism (Sections 7.1–7.8)

— End of Question Paper —